

EDS223 - HW1

Map making practice

Sophia Soriano

Map 1: Western US - Air Toxins (Lifetime Inhalation) Cancer Risk US Percentile

```
# Load packages
library(tidyverse)
library(sf)
library(here)
library(tmap)
```

```
# DATA PROCESSING CODE
```

```
# Load geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI"), as_sf = TRUE)
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Filter to the west US states - maintain county borders
west_us <- ejscreen %>%
  dplyr::filter(ST_ABBREV %in% c("CA", "OR", "WA", "NV")) # "ID", "AZ", "UT"
```

```

# Filter to state borders only
west_us_states <- aggregate(west_us, by = list(west_us$ST_ABBREV), FUN = mean)

# Calculate average values for all variables within west US counties
west_us_counties <- aggregate(west_us, by = list(west_us$CNTY_NAME), FUN = mean)

# Create ENVIRONMENTAL map - west US, air toxin cancer risk (US percentile)
# Map west US counties as polygons - base layer
tm_shape(west_us_counties) +
  tm_polygons(# Use P_CANCER values to color polygons
    fill = "P_CANCER",
    # Define color scale
    fill.scale = tm_scale(values = c("#ade4f8", "#e3271aae")),
    fill.legend = tm_legend(title = "Air Toxins Cancer Risk\n(US Percentile)",
      # Specify legend layout
      position = tm_pos_out("right")),
    # Set line width for county borders
    lwd = 0.5
  ) +
# Map west US states as borders - 2nd layer
tm_shape(west_us_states) +
  tm_borders(# Set line color for state borders
    col = "black",
    # Set line width for state borders
    lwd = 2) +
# Add compass to bottom left
tm_compass(position = c("left", "bottom")) +
# Add scalebar to bottom left
tm_scalebar(position = c("left", "bottom")) +
# Specify additional layout options
tm_layout(main.title = "Western US: Lifetime Inhalation Cancer Risk per County",
  # Slightly increase title size (ratio)
  main.title.size = 1.5,
  # Add title frame box
  main.title.frame = TRUE,
  # Set aspect ratio to adjust dynamically with display area (not static)
  asp = 0,
  # Increase inner margins to keep compass and scale bar in background area
  inner.margins = c(0.1, 0.05, 0.05, 0.05),
  # Specify legend title size
  legend.title.size = 1.1,
  # Specify legend text size

```

```

legend.text.size = 0.9,
# Set background color
bg.color = "#f3f6f8",
# Set font family
text.fontfamily = "sans"
)

```

Western US: Lifetime Inhalation Cancer Risk per County

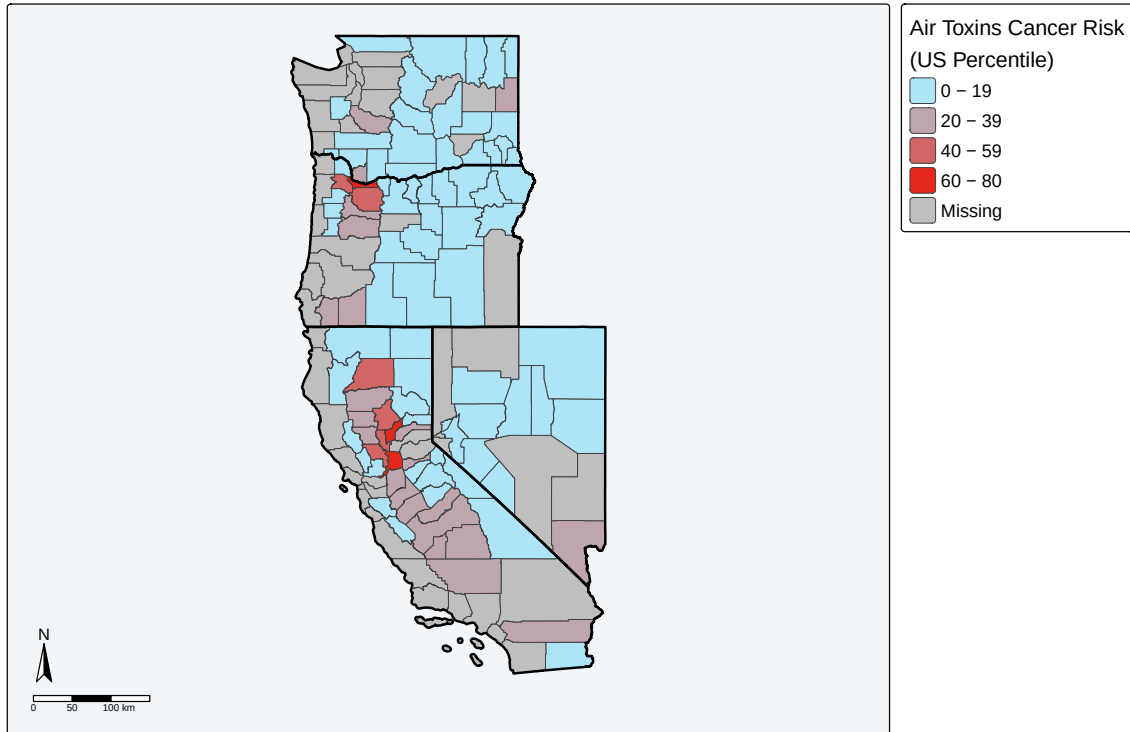


Figure 1: This first map of counties in the western US (CA, NV, OR, WA) is colored by the lifetime air toxin inhalation cancer risk US percentile (see Discussion below for definition). Blue equates to counties in minimal risk percentiles, while red signals counties in higher risk percentiles. Grey denotes no data available.

Map 2: Western US - Percentage People of Color by County US Percentile

```

# Create DEMOGRAPHIC map - west US, people of color % (US percentile)
# Map west US counties as polygons - base layer
tm_shape(west_us_counties) +
  tm_polygons(# Use P_PEOPCOLORPCT values to color polygons
    fill = "P_PEOPCOLORPCT",
    # Define color scale
    fill.scale = tm_scale(values = c("#f8fabb", "#e3271aae")),
    fill.legend = tm_legend(title = "Percentage People of Color\n(US Percentile)",
      # Specify legend layout
      position = tm_pos_out("right")),
    # Set line width for county borders
    lwd = 0.5) +
# Map west US states as borders - 2nd layer
tm_shape(west_us_states) +
  tm_borders(# Set line color for state borders
    col = "black",
    # Set line width for state borders
    lwd = 2) +
# Add compass to bottom left
tm_compass(position = c("left", "bottom")) +
# Add scalebar to bottom left
tm_scalebar(position = c("left", "bottom")) +
# Specify additional layout options
tm_layout(main.title = "Western US: People of Color per County",
  # Slightly increase title size (ratio)
  main.title.size = 1.5,
  # Add title frame box
  main.title.frame = TRUE,
  # Set aspect ratio to adjust dynamically with display area (not static)
  asp = 0,
  # Increase inner margins to keep compass and scale bar in background area
  inner.margins = c(0.1, 0.05, 0.05, 0.05),
  # Specify legend title size
  legend.title.size = 1.1,
  # Specify legend text size
  legend.text.size = 0.9,
  # Set background color
  bg.color = "#f3f6f8",
  # Set font family
  text.fontfamily = "sans"
)

```

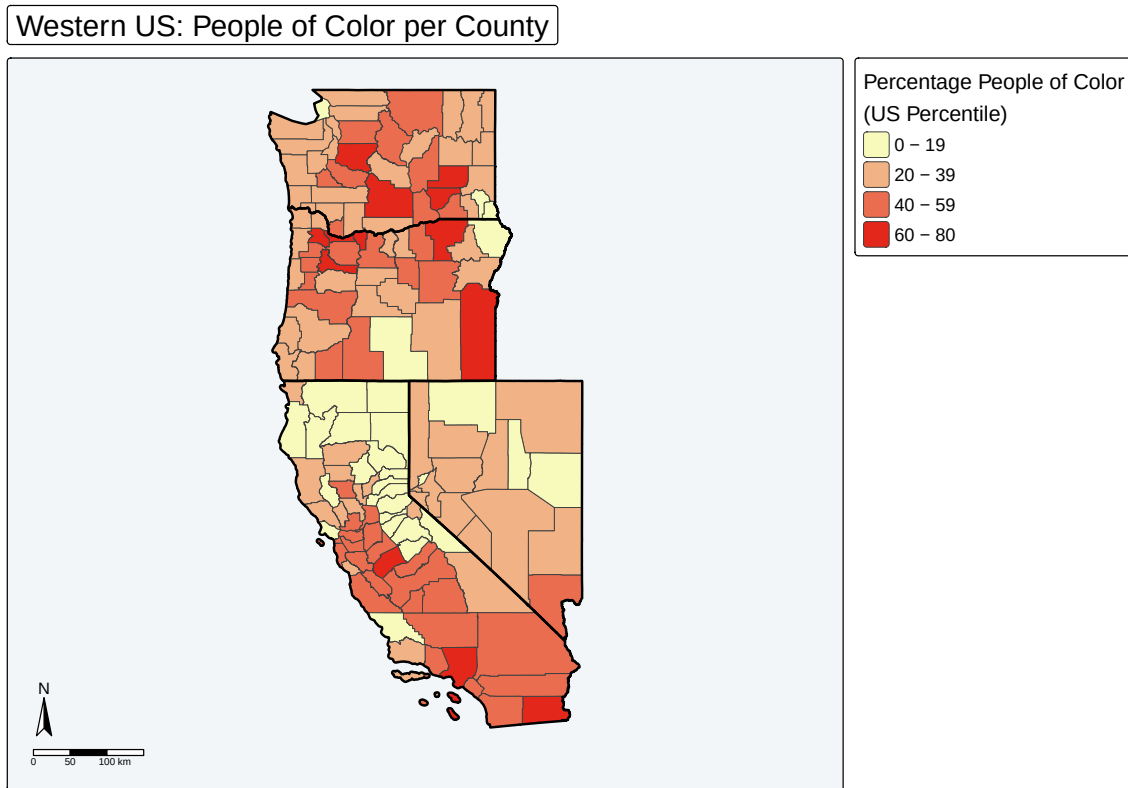


Figure 2: This second map of counties in the western US (CA, NV, OR, WA) is colored by the US percentile of the percentage people of color (see Discussion for definition). Blue equates to counties in a lower percentile, while red signals counties in a higher percentile. Grey denotes no data available.

Discussion

As part of my previous graduate program in bioinformatics at University of Oregon, I worked as a Computational Biologist in the cancer proteomics group at the Oregon Health & Science University (OHSU) Cancer Early Detection Advanced Research Center (CEDAR). After surveying the types of data collected through the EJScreen: Environmental Justice Screening and Mapping Tool and reading the column descriptions, I noticed the Air Toxins Cancer Risk columns and decided to see if any of the demographic percentiles had a similar pattern to the cancer risk percentiles in the Western US (CA, NV, OR, WA). Percentile appeared to be a more appropriate unit to compare an environmental variable with a demographic variable. The air toxins cancer risk is defined in the documentation as “the estimated lifetime inhalation cancer risk from the analyzed carcinogens in ambient outdoor air, as provided by the 2019 AirToxScreen.” This variable did not include skin cancer-related data, and “lifetime” was

defined as 70 years. The people of color measurement was determined by the percentage of people identifying as any category other than “non-Hispanic white-alone individuals.”

Above in **Figure 1** and **Figure 2** are maps of the Western US, the first mapping the US percentile of air toxin (inhalation) cancer risk (P_CANCER) by county, and the second mapping the US percentile of the percentage people of color (P_PEOPCOLORPCT) by county. The following areas have counties in higher percentiles for both air toxin cancer risk and percentage people of color compared to other mapped areas: south and central valley areas of California, southern Nevada, and western Oregon. While no statistics have been evaluated here, the fact that the counties in these areas fall in both of these higher percentile categories signals a potential for an environmental justice issue related to racial bias in air toxin monitoring and control.

Data

The data used for this assignment is from the EJScreen: Environmental Justice Screening and Mapping Tool, which no longer exists online. However, our data can be accessed through the following link: <https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view>

References

[EDS223 - UCSB MEDS 2027](#)